

ECE 417/598 Midterm 2 2024

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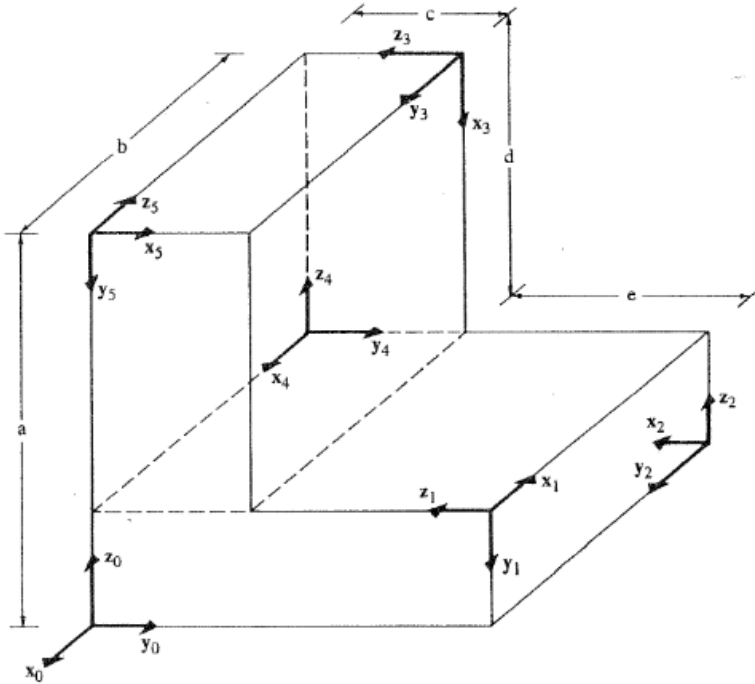
(1) Student name:

Student email:

About the exam

1. There are total 5 problems. You must attempt all 5.
2. Maximum marks: 50.
3. Maximum time allotted: 50 min
4. Calculators are allowed.
5. One 8x11 in cheat sheet (both-sides) is allowed.

Problem 1 Find the 4×4 transformation matrix 0T_2 that transforms coordinates from coordinate frame 2 to coordinate frame 0 (5 marks).



Solution: I am going to write transform from frame 1 to frame 0. 0T_2

$${}^0T_2 = \begin{bmatrix} 0 & 1 & 0 & -b \\ -1 & 0 & 0 & c+e \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

The rotation matrix is obtained by writing the basis vectors of the destination coordinate system in the source coordinate system.

Problem 2 In your own words, prove using trigonometry that the 2D rotation matrix is given by (10 marks)

$$R = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix}. \quad (2)$$

Solution:

We can express original coordinates (x, y) in polar coordinates,

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} r \cos(\alpha) \\ r \sin(\alpha) \end{bmatrix}, \quad (3)$$

where $r = \sqrt{x^2 + y^2}$ is the length of the vector and $\alpha = \arctan2(y, x)$ is the angle between the vector and X-axis.

The rotated points (x', y') have the same length r but a different angle

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} r \cos(\alpha + \alpha) \\ r \sin(\alpha + \alpha) \end{bmatrix}. \quad (4)$$

Using the trigonometric identities we can write,

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} r \cos(\alpha) \cos(\alpha) - r \sin(\alpha) \sin(\alpha) \\ r \sin(\alpha) \cos(\alpha) + r \cos(\alpha) \sin(\alpha) \end{bmatrix}. \quad (5)$$

Substituting $r \cos(\alpha) = x$ and $r \sin(\alpha) = y$, we get

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \cos(\alpha) - y \sin(\alpha) \\ x \sin(\alpha) + y \cos(\alpha) \end{bmatrix}. \quad (6)$$

Writing the right hand side (RHS) as a matrix vector product,

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \underbrace{\begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & +\cos(\alpha) \end{bmatrix}}_R \begin{bmatrix} x \\ y \end{bmatrix}. \quad (7)$$

The matrix being multiplied here is the 2D rotation matrix.

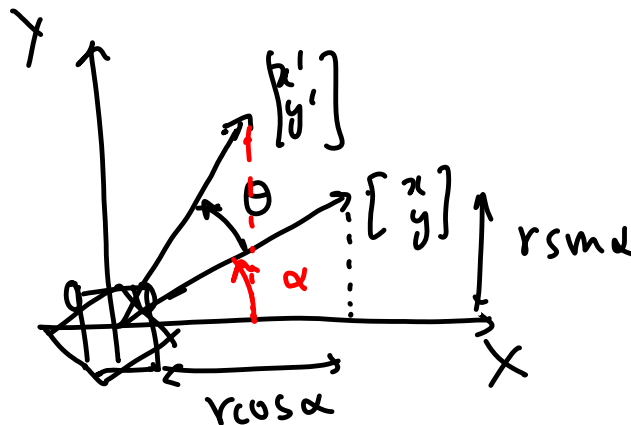
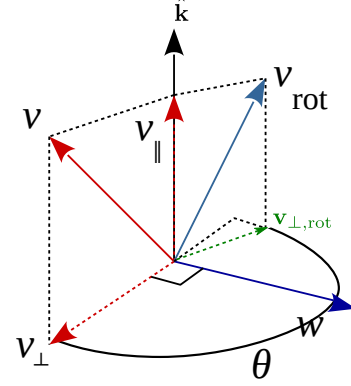


Figure 1: Rotation of points $[x, y]$ to $[x', y']$ by an angle α

Problem 3 (Rodrigues formula) In the figure below, we rotate a point $\mathbf{v}$ around axis unit-vector $\hat{\mathbf{k}}$ by an angle θ . A unit vector $\hat{\mathbf{w}}$ is perpendicular to the both $\mathbf{v}$ and $\hat{\mathbf{k}}$. Another vector $\mathbf{v}_\perp$ is the projection of $\mathbf{v}$ onto a plane that is perpendicular to $\hat{\mathbf{k}}$. Note that $\mathbf{v}_\perp$ is perpendicular to both $\hat{\mathbf{w}}$ and $\hat{\mathbf{k}}$ (10 marks).



- write the unit-vector $\hat{\mathbf{w}}$ in terms of $\mathbf{v}$ and $\hat{\mathbf{k}}$.
- Then write the vector (including the correct magnitude) $\mathbf{v}_\perp$ in terms of $\mathbf{v}$ and $\hat{\mathbf{k}}$ (5 marks).

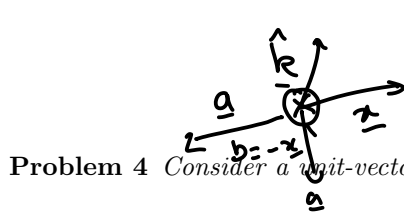
Solution:

$$\hat{\mathbf{w}} = \frac{\hat{\mathbf{k}} \times \mathbf{v}}{\|\hat{\mathbf{k}} \times \mathbf{v}\|} \quad (8)$$

Note that $\hat{\mathbf{k}} \times \mathbf{v}$ has the magnitude $\|\hat{\mathbf{k}} \times \mathbf{v}\| = |\hat{\mathbf{k}}||\mathbf{v}|\sin(\alpha) = |\mathbf{v}|\sin(\alpha)$ where α is the angle between $\mathbf{v}$ and $\hat{\mathbf{k}}$.

$$\mathbf{v}_\perp = -\hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times \mathbf{v}) \quad (9)$$

The magnitude of $\mathbf{v}_\perp$ is same as RHS because both $|\mathbf{v}_\perp| = |\mathbf{v}|\sin(\alpha)$ and $|\hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times \mathbf{v})| = |\mathbf{v}||\hat{\mathbf{k}}|^2 \sin(\alpha) \sin(90^\circ) = |\mathbf{v}|\sin(\alpha)$ where α is the angle between $\mathbf{v}$ and $\hat{\mathbf{k}}$.



Problem 4 Consider a unit-vector $\hat{\mathbf{k}} = (k_x, k_y, k_z)^\top$ and the corresponding crossproduct matrix,

$$K = [\hat{\mathbf{k}}]_\times = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix}.$$

$$(\hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times \mathbf{x}))) = -(\hat{\mathbf{k}} \times \mathbf{x})$$

$$K^3 = -K \quad (10)$$

Using the geometry of the cross-product to prove that $K^3 = -K$. Hint on next page. (5 marks)

Solution: Geometric method:

Choose any vector $\mathbf{x} \neq \gamma \hat{\mathbf{k}}$. Draw the following three vectors: (1) $\mathbf{a} = \hat{\mathbf{k}} \times \mathbf{x}$, (2) $\mathbf{b} = \hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times \mathbf{x})$, (3) $\mathbf{c} = \hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times \mathbf{x}))$. Compare vectors $\mathbf{c}$ and $\mathbf{a}$. Note that the direction of $\mathbf{c}$ is opposite to $\mathbf{a}$. In other words, $\mathbf{c} = -|\beta| \mathbf{a}$ where $|\beta|$ is some scalar.

What is magnitude of $\mathbf{c}$ and $\mathbf{a}$?

$$|\mathbf{c}| = |\hat{\mathbf{k}}|^3 |\mathbf{x}| \sin(\alpha) \sin(90) \sin(90) = |\mathbf{x}| \sin(\alpha) \quad (11)$$

$$|\mathbf{a}| = |\hat{\mathbf{k}}| |\mathbf{x}| \sin(\alpha) = |\mathbf{x}| \sin(\alpha), \quad (12)$$

where α is the angle between $\hat{\mathbf{k}}$ and $\mathbf{x}$. Thus $|\mathbf{c}| = |\mathbf{a}|$.

Because $|\mathbf{c}| = |\mathbf{a}|$ and $\mathbf{c} = -|\beta| \mathbf{a}$, therefore $\mathbf{c} = -\mathbf{a}$, or

$$\hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times \mathbf{x})) = -\hat{\mathbf{k}} \times \mathbf{x} \quad (13)$$

$$\implies K^3 \mathbf{x} = -K \mathbf{x}. \quad (14)$$

Since $K^3 \mathbf{x} = -K \mathbf{x}$ for any $\mathbf{x} \neq \gamma \hat{\mathbf{k}}$, hence $K^3 = -K$. (When $\mathbf{x} = \gamma \hat{\mathbf{k}}$, then $K \mathbf{x} = \mathbf{0} = K^3 \mathbf{x}$).

Proof by algebra

This was not allowed because I asked to use geometry of the cross-product. But it is possible to prove this using algebra as well.

$$K^2 = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix} \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix} = \begin{bmatrix} -(k_z^2 + k_y^2) & k_y k_x & k_z k_x \\ k_x k_y & -(k_x^2 + k_z^2) & k_z k_y \\ k_x k_z & k_y k_z & -(k_y^2 + k_x^2) \end{bmatrix} \quad (15)$$

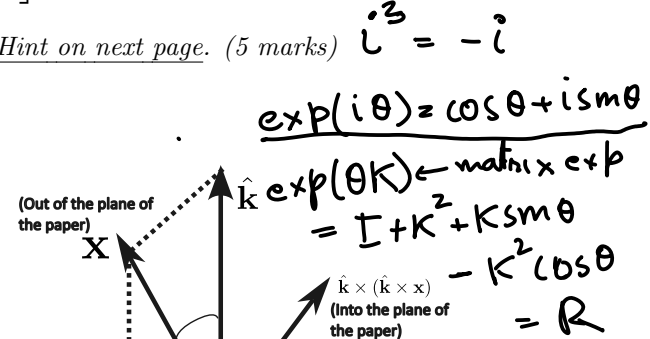
$$\implies K^3 = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix} \begin{bmatrix} -(k_z^2 + k_y^2) & k_y k_x & k_z k_x \\ k_x k_y & -(k_x^2 + k_z^2) & k_z k_y \\ k_x k_z & k_y k_z & -(k_y^2 + k_x^2) \end{bmatrix} \quad (16)$$

$$= \begin{bmatrix} -k_z k_y k_x + k_y k_x k_z, & k_z(k_x^2 + k_z^2) + k_y^2 k_z, & -k_z^2 k_y - k_y(k_y^2 + k_x^2) \\ -k_z(k_z^2 + k_y^2) - k_x^2 k_z, & -k_z k_y k_x + k_y k_x k_z, & k_z^2 k_x + k_x(k_y^2 + k_z^2) \\ k_y(k_z^2 + k_y^2) + k_x^2 k_y, & -k_y^2 k_x - k_x(k_x^2 + k_z^2), & -k_z k_y k_x + k_y k_x k_z \end{bmatrix} \quad (17)$$

$$= \begin{bmatrix} 0, & k_z(k_x^2 + k_z^2 + k_y^2), & -k_y(k_z^2 + k_y^2 + k_x^2) \\ -k_z(k_z^2 + k_y^2 + k_x^2), & 0, & k_x(k_z^2 + k_y^2 + k_x^2) \\ k_y(k_z^2 + k_y^2 + k_x^2), & -k_x(k_z^2 + k_y^2 + k_x^2), & 0 \end{bmatrix}. \quad (18)$$

Use the fact, $k_x^2 + k_y^2 + k_z^2 = \|\hat{\mathbf{k}}\|^2 = 1$.

$$K^3 = \begin{bmatrix} 0 & k_z & -k_y \\ -k_z & 0 & k_x \\ k_y & -k_x & 0 \end{bmatrix} = - \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix} = -K \quad (19)$$



$$\frac{d}{d\theta} R(\theta, \hat{k}) = \frac{d}{d\theta} \exp(\theta K) = \underline{\exp(\theta K)} K$$

$$= R K$$

$$\frac{d}{dx} \exp(x) = \exp(x)$$

$$\exp(i\theta) = \sin\theta + i\cos\theta \quad \leftarrow \text{how to you prove this}$$

↓
Series form

$$\exp(i\theta) = 1 + \frac{i\theta}{1!} + \frac{i^2\theta^2}{2!} + \frac{i^3\theta^3}{3!} + \dots \dots \infty$$

$$= \underbrace{\left(1 + \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \dots + \infty\right)}_{\sin\theta} + i \underbrace{\left(\frac{\theta}{1!} + \frac{\theta^3}{3!} + \dots + \infty\right)}_{\cos\theta}$$

$$\exp(\theta K) = I + \frac{\theta K}{1!} + \frac{\theta^2 K^2}{2!} + \dots$$

(Hint for Problem 4: Choose any vector $\mathbf{x} \neq \gamma \hat{\mathbf{k}}$. Draw the following three vectors: (1) $\mathbf{a} = \hat{\mathbf{k}} \times \mathbf{x}$, (2) $\mathbf{b} = \hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times \mathbf{x})$, (3) $\mathbf{c} = \hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times \mathbf{x}))$. Compare vectors $\mathbf{c}$ and $\mathbf{a}$. What do you observe?)

Problem 5 Derive the rotation matrix corresponding to the Euler angles (α, β, γ) representation $R = R_x(\alpha)R_y(\beta)R_z(\gamma)$. Also derive an expression to convert the rotation matrix back to Euler angles. (20 marks).

Solution:

$$R = R_x(\alpha)R_y(\beta)R_z(\gamma) \quad (20)$$

$$\Rightarrow \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\alpha & -s_\alpha \\ 0 & s_\alpha & c_\alpha \end{bmatrix} \begin{bmatrix} c_\beta & 0 & s_\beta \\ 0 & 1 & 0 \\ -s_\beta & 0 & c_\beta \end{bmatrix} \begin{bmatrix} c_\gamma & -s_\gamma & 0 \\ s_\gamma & c_\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (21)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\alpha & -s_\alpha \\ 0 & s_\alpha & c_\alpha \end{bmatrix} \begin{bmatrix} c_\beta c_\gamma & -c_\beta s_\gamma & s_\beta \\ s_\gamma & c_\gamma & 0 \\ -s_\beta c_\gamma & s_\beta s_\gamma & c_\beta \end{bmatrix} \quad (22)$$

$$\Rightarrow \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} c_\beta c_\gamma & -c_\beta s_\gamma & s_\beta \\ c_\alpha s_\gamma + s_\alpha s_\beta c_\gamma & c_\alpha c_\gamma - s_\alpha s_\beta s_\gamma & -s_\alpha c_\beta \\ s_\alpha s_\gamma - c_\alpha s_\beta c_\gamma & s_\alpha c_\gamma + c_\alpha s_\beta s_\gamma & c_\alpha c_\beta \end{bmatrix} \quad (23)$$

$$\beta = \sin^{-1}(-r_{13}) \in [-\pi/2, \pi/2] \quad (24)$$

$$\gamma = \arctan2(-r_{12}, r_{11}) \quad (25)$$

$$\alpha = \arctan2(-r_{23}, r_{33}) \quad (26)$$